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Name of the Programme:	**B.Sc(H)**	
Title of the Course:	**Data Science**	
Course code	**CS2235**	
Semester:	**III**	
Names of the Course Instructor(s)	**Prof. Muralidhar Billa** **Prof. Thirukumaran Subbiramani**	
Course credit	**No. of hours per week (1 + 1 + 2)**	**Total no. of Teaching hours**
3	**4**	**60**

Capstone Project Guidelines

1. Objective

The Capstone Project aims to provide students with hands-on experience in applying data science concepts such as data collection, cleaning, transformation, model building, and evaluation to solve real-world problems. Each team will complete an end-to-end data science workflow using Python and relevant libraries.

2. Team Structure

- Each team should consist of **three** / **four members and divide the work accordingly**.
- Each team must nominate the following roles:
    - **Team Leader:** Coordinates the overall project and communication with the instructor.
    - **Documentation :** Prepares and finalizes the written report.
    - **Coding :** Integrates code from all members and ensures version control.
    - **Presentation :** Prepares and delivers the final presentation.

3. Project Workflow

Phase	Description	Deliverables
Phase 1: Project Kickoff	Topic selection, dataset identification, and framing of problem statement.	Project proposal (title, dataset, objective).
Phase 2: Data Collection and Cleaning	Acquire dataset, handle missing values, and perform data cleaning.	Cleaned dataset and notebook.
Phase 3:	Perform data transformation, encoding, scaling, and feature selection.	ETL notebook with documentation.

ETL and Feature Engineering		
Phase 4: Model Implementation	Select suitable machine learning algorithm (Regression, Classification, or Clustering) and train the model.	Model code and results.
Phase 5: Evaluation and Deployment	Evaluate model performance, visualize results, and develop a simple deployment (Flask/Streamlit/Jupyter).	Working demo or local deployment.
Phase 6: Presentation and Viva	Submit final report and present findings.	Report, PPT, and viva presentation.

4. Dataset and Problem Identification

Students may use datasets from public sources such as:

- Kaggle
- UCI Machine Learning Repository
- data.gov.in
- Open-source CSV datasets

Identifying the Type of Problem

Type	Output Variable	Example	Common Algorithms
Regression	Continuous numerical value	Predicting house price, rainfall amount	Linear Regression, Decision Tree Regressor
Classification	Categorical/discrete label	Spam detection, disease prediction	Logistic Regression, KNN, Random Forest
Clustering	No predefined label	Customer segmentation, grouping students	K-Means, Hierarchical Clustering

Note:

- If the target/output variable is *numerical*, it is a regression problem.
- If the target/output variable is *categorical*, it is a classification problem.
- If there is *no target variable*, it is a clustering (unsupervised) problem.

5. Rules and Regulations

1. Each team must submit a project proposal and obtain approval before starting work.
2. All members should contribute equally; the team leader ensures fair participation.
3. The project must be implemented using **Python (preferably Jupyter Notebook)**.

4. Use of external AI tools for code/report generation is not permitted.
5. Late submissions will attract marks deduction.
6. Plagiarism above 25% will lead to disqualification.
7. Every student must be able to explain their project during the viva.
8. Teams are encouraged to maintain backup copies or use GitHub/Google Drive for version control.

6. Deliverables Checklist

Deliverable	**Description**
Project Proposal	Title, team details, dataset source, problem statement, and expected output.
Jupyter Notebook	Complete and well-documented code covering data cleaning to model evaluation.
Report	PDF report (8–10 pages) covering introduction, methodology, results, and conclusionwith 2500 words.
Presentation	6–8 slides summarizing the project workflow and findings.
Demo	Working project demonstration or application interface (optional but encouraged).

7. Evaluation Scheme (Total: 25 Marks)

Component	**Description**	**Marks**
Implementation	Complete workflow with innovative or optimized implementation.	5
Project presentation	Modular, well-documented code with clarity and explaining the project.	5
Report Submission	Comprehensive and well-organized report with proper formatting.	5
Viva	Confident and accurate explanation of project methodology and results.	10

8. Recommended Tools and Libraries

- **Python Libraries:** Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn
- **Deployment Tools (Optional):** Flask, Streamlit
- **Collaboration Tools:** Google Drive, GitHub

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9. Suggested Project Topics (But Don't choose them; analyze and pick another dataset)

1. Predicting Student Performance (Regression)
2. Iris Flower Classification (Classification)
3. Customer Segmentation (Clustering)
4. Movie Rating Prediction (Regression)
5. Fake News Detection (Classification)
6. Crop Recommendation System (Classification)
7. Air Quality Index Prediction (Regression)
8. Online Retail Customer Segmentation (Clustering)

10. Suggested Report Format

1. Title Page
2. Abstract
3. Introduction and Problem Statement
4. Related Work / Background
5. Dataset Description
6. Methodology (Data Cleaning, ETL, Model Building)
7. Results and Evaluation
8. Discussion
9. Conclusion and Future Work
10. References